

Education

University of California, San Diego | San Diego, CA SEP 2024 - DEC 2025
M.S. in Computer Science (AI focus) GPA: 3.90

Relevant Coursework: Deep Learning, Machine Learning, NLP, ML Systems, Artificial Intelligence, Recommender Systems, Computer Arch
University of California, Berkeley | Berkeley, CA AUG 2020 - MAY 2024
B.A. in Computer Science & B.A. in Applied Math GPA: 3.90

Professional Experience

Member of Technical Staff | Axilon DEC 2025 - PRESENT
• Ontologizing your factory; deploying scalable agentic harnesses and the AI-native context layer for the industrial world

Applied ML Engineer (Contract) | Outerproduct FEB 2026 - APR 2026
• Distributed training and steering of LLMs (Mistral, Llama) using recursive feature machines to improve performance & interpretability

Machine Learning Engineer Intern | Qualcomm JUN 2025 - SEP 2025
• Architect multimodal RAG system through Langchain, using ONNX-based CLIP embeddings, FAISS vector store, Qwen2.5VL
• Deploy system on Qualcomm-native CPU machine, benchmarking against CUDA host on Haystacks using LLM-as-a-judge
• Cross-compile Llama.cpp for Adreno GPU and quantize Qwen2.5 VL model to Q4_0, achieving over 10x speedup from baseline CPU
• Extensively profile and fix CLIP projection encoding bottleneck in Llama.cpp C++ binaries, achieving 5x speedup from naive solution

Teaching Assistant | CSE 234: ML Sys, CSE 253: ML for Music, CSE 258: Recommenders @ UCSD DEC 2024 - DEC 2025
• Constructed 3 brand-new assignments for students covering mat-mul optimization, backprop graph construction, speculative decoding
• Hosted 50+ office hours, proctor exams, hold discussion sections, maintain Ruby website under Prof. Hao Zhang & Prof. Julian McAuley

Research Intern | MUSAIC, Hao AI Lab @ UCSD APR 2024 - DEC 2025
• Explore interpretability of music and guided real-time music generation through recursive feature machines and sparse autoencoders
• Configured new CLLM models to be compatible with qLoRA fine-tuning on Jacobi trajectories, decreasing memory usage over 3x

Machine Learning Engineer Intern | Accretional JUN 2024 - SEP 2024
• Designed and implement RAG vector embedding model to ground LLM with respect to cloud infrastructure
• Fully constructed VSCode application in Svelte and TypeScript, creating tool-calling workflow in the agentic AI pipeline
• Created command-line tool in Golang for indexing and vectorization of files, semantic & lexical search of keywords

Software Engineer Intern | Hinge Health MAY 2023 - AUG 2023
• Designed 5+ endpoints and event consumers in TypeScript and Ruby on Rails, restructuring and building new GraphQL queries
• Presented solutions for over 10 UI/UX features in React, linked repos with RabbitMQ event emitting, deployed in Kubernetes

Full Stack Intern | Cisco Meraki MAY 2022 - AUG 2022
• Spearheaded a full stack project to create uplink ping stat graphs, helping 500,000 customers visualize wireless trends over time
• Collected ping stats data from over 4 million wireless internet nodes, using Scala scripts to build a grabber and Jenkins to test

Research

Steering Autoregressive Music Generation with RFMs (ICLR 2026, first author) JAN 2026
• Interpretable classification of music using recursive feature machines (based on kernel ridge regression & AGOP) as classifiers
• Inject direction vectors into LLM's logits layer to steer LLM towards a musically interpretable concept (e.g. major vs. minor)

VideoScience-Bench (ECCV 2026), an auto-eval physics reasoning benchmark for video generation DEC 2025

CoT Reasoning with Sparse Autoencoders, clustering on SAE-based token rep for LLM interpretability FEB 2025

Skills

Python, PyTorch, TypeScript, Machine Learning, AI & Agentic Systems, System Design, AWS, Docker, Kubernetes, Git, Deployment Workflows, SQL, GraphQL, Go, React, TensorFlow, RAG, LLM Evaluation, Knowledge Graphs, C/C++, Java, Ruby, Unix, Scala